



MISO DPP 2020 Cycle 1 South Area Study Phase I Final Report

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**MISO
720 City Center
Drive Carmel
Indiana 46032**
<http://www.misoenergy.org>

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1. Executive Summary

This report presents the results of a System Impact Study performed to evaluate the interconnection of the generators in the DPP 2020 Cycle 1 South Area Phase I. The study was performed under the direction of MISO and reviewed by an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Entergy, Cooperative Energy, CLECO, AECC, AECL, Southern Company, PowerSouth, SPP, and TVA.

1.1. Project List

The original interconnection requests for DPP 2020 Cycle 1 South Area had a total of 81 generation projects. Projects J1505, J1506, J1507, J1538, J1541, J1545, J1551, J1574, J1584, J1595, J1640, J1648, J1674, J1685, J1698, J1787, and J1788 withdrew prior to the start of DPP Phase I. Therefore, there are 64 generation projects with a combined nameplate rating of 8849.8 MW of Energy Resource Interconnection Service (ERIS) & 8957.8 MW of Network Resource Interconnection Service (NRIS). The detailed list of South Area DPP projects is shown below in Table 1, and the Phase 1 study was kicked off on December 22nd 2020.

Table 1: List of DPP 2020 Cycle 1 South Area Phase I Projects

Project	Fuel Type	Transmission Owner	County	State	Service Requested	MW	Point of Interconnection
J1669	Solar	Entergy Louisiana LLC	Jefferson Davis	LA	ERIS	165	LC Bulk to Henning 138kV Line (Circuit 28)
J1671	Solar	Entergy Texas Inc	Walker	TX	ERIS	150	Rocky Creek to Lewis Creek 230kV
J1794	Solar	Entergy Louisiana LLC	Acadia	LA	ERIS	150	Richard to Scott 138kV line
J1795	Solar	Entergy Louisiana LLC	Acadia	LA	ERIS	100	Bayou Cove to Richard 138kV (Circuit 258)
J1499	Battery Storage	Entergy Arkansas LLC	Union	AR	NRIS	50	El Dorado East - Sterlington 115 kV
J1500	Solar	Cooperative Energy	George	MS	NRIS	150	SE Greene Co 230kV substation
J1509	Solar	Entergy Mississippi LLC	Leflore	MS	NRIS	100	Greenwood - Pickens 115 kV
J1514	Solar	Entergy Arkansas LLC	Mississippi	AR	NRIS	100	Driver 230kV
J1523	Battery Storage	Entergy Mississippi LLC	Leflore	MS	NRIS	50	Greenwood - Pickens 115 kV
J1539	Solar	Entergy Louisiana LLC	West Baton Rouge	LA	NRIS	146	J683 POI 138 kV
J1542	Solar	Entergy Louisiana LLC	St. James	LA	NRIS	200	Thibodaux - Vacherie 230kV
J1558	Solar	Entergy Arkansas LLC	Pope	AR	NRIS	200	Moreland - St. Vincent 161kV line
J1559	Solar	Entergy Arkansas LLC	Conway	AR	NRIS	200	Morrilton East Substation (161kV bus)
J1560	Solar	CLECO Corporation	Iberia	LA	NRIS	200	Patoutville Substation (138kV bus)
J1561	Solar	Entergy Louisiana LLC	Acadia	LA	NRIS	150	Richard 138kV
J1562	Solar	Entergy Arkansas LLC	Mississippi	AR	NRIS	200	Blytheville I-55 to AECC Blytheville North 161kV
J1564	Solar	Entergy Louisiana	Washington	LA	NRIS	100	Talisheek - Bogalusa 230

		LLC					kV Line
J1576	Solar	Entergy Louisiana LLC	Washington	LA	NRIS	200	Angie to Adams Creek 230 kV
J1577	Solar	Entergy Arkansas LLC	Grant	AR	NRIS	100	Hardin West SS (Endpoint A) to Sheridan (Endpoint B) 115 kV
J1583	Solar	Entergy Louisiana LLC	Iberville	LA	NRIS	200	Donaldsonville - Evergreen 230kV
J1592	Solar	Entergy Arkansas LLC	Pope	AR	NRIS	200	MORRILTON EAST - Atkins 161 kV
J1598	Solar	Entergy Louisiana LLC	Calcasieu	LA	NRIS	325	Solac - Chalkley 230kV
J1599	Hybrid	CLECO Corporation	Iberia	LA	NRIS	100	Morbihan - Hopkins 138kV
J1602	Battery Storage	Entergy Louisiana LLC	Ouachita	LA	NRIS	25	Rilla 115kV substation (Same as J1239)
J1607	Battery Storage	Entergy Arkansas LLC	Grant	AR	NRIS	0	Hardin West SS to Sheridan 115 kV
J1612	Solar	Entergy Mississippi LLC	Union	AR	NRIS	100	El Dorado East - Sterlington 115 kV
J1618	Solar	Entergy Louisiana LLC	Ascension	LA	NRIS	200	ST. JAMES - WELCOME 230KV
J1630	Solar	Entergy Mississippi LLC	Madison	MS	NRIS	150	Gerald Andrus-Tinnin Road 230 kV
J1633	Battery Storage	CLECO Corporation	Beauregard	LA	NRIS	0	Cooper Switching - Penton Road 230 kV
J1634	Solar	CLECO Corporation	Beauregard	LA	NRIS	100	COOPER SWITCHING - PENTON ROAD 230 KV
J1644	Solar	Entergy Louisiana LLC	St. Landry	LA	NRIS	150	Champagne - Plaisance
J1668	Solar	Entergy Louisiana LLC	Iberville	LA	NRIS	194	Donaldsonville - Evergreen 230 kV
J1670	Solar	Entergy Arkansas LLC	Mississippi	AR	NRIS	50	Crooked Lake 161kV at J586 POI
J1672	Solar	Entergy Mississippi LLC	Hinds	MS	NRIS	150	Tinnin Road 230kV
J1673	Solar	CLECO Corporation	Sabine	LA	NRIS	200	Fisher to Mansfield Compressor 230 kV
J1686	Hybrid	CLECO Corporation	Evangeline	LA	NRIS	100	Coughlin - Plaisance 138 kV
J1709	Battery Storage	Entergy Texas Inc	Grimes	TX	NRIS	100	138 kV Grimes
J1710	Solar	Entergy Arkansas LLC	Lee	AR	NRIS	300	Entergy Keo-Memphis 500kV Line Tap
J1722	Solar	Entergy Arkansas LLC	Lonoke	AR	NRIS	100	Galloway - Remigton 115kV Transmission Line
J1723	Battery Storage	Entergy Arkansas LLC	Cross	AR	NRIS	10	Wynne East to Wynne South 161kV line (J1125 sub)
J1747	Solar	Entergy Mississippi LLC	Madison	MS	NRIS	50	J830 POI
J1748	Solar	Entergy Mississippi LLC	DeSoto	MS	NRIS	100	Twinkletown 230 kV
J1749	Hybrid	Entergy Louisiana	West Baton	LA	NRIS	50	Addis - Big Cajun 230kV

		LLC	Rouge				
J1759	Battery Storage	CLECO Corporation	Sabine	LA	NRIS	0	Fisher to Mansfield Compressor 230 kV
J1760	Solar	Entergy Texas Inc	Liberty	TX	NRIS	250	Timberland 230kV
J1789	Solar	Entergy Mississippi LLC	Monroe	AR	NRIS	5.99	Brinkley East to Clarendon 115 kV
J1790	Battery Storage	Entergy Arkansas LLC	Ashley	AR	NRIS	0	Crossett - Monticello East 115 kV Line
J1791	Battery Storage	Entergy Mississippi LLC	Madison	MS	NRIS	0	Gerald Andrus-Tinnin Road 230 kV
J1796	Solar	CLECO Corporation	Evangeline	LA	NRIS	150	Pine Prairie to Coughlin 138 kV
J1797	Solar	Entergy Louisiana LLC	St. John the Baptist	LA	NRIS	100	Waterford 230kV
J1811	Solar	Entergy Arkansas LLC	Independence	AR	NRIS	150	Independence 161 kV
J1812	Wind	Entergy Texas Inc	Chambers	TX	NRIS	294.25	Stowell 230 kV
J1813	Gas	Entergy Mississippi LLC	DeSoto	MS	NRIS	107	Entergy Freeport 230 kV Substation
J1816	Solar	CLECO Corporation	Cross	AR	NRIS	135	Fisher (Endpoint A) to Cherry Valley (Endpoint B) 161 kV line
J1818	Solar	Lafayette Utilities System	Lafayette	LA	NRIS	80	Northeast (Moss) 69kV
J1819	Battery Storage	Entergy Arkansas LLC	Lee	AR	NRIS	0	Entergy Keo - West Memphis 500kV Line Tap
J1820	Solar	Entergy Arkansas LLC	Lee	AR	NRIS	100	Entergy Keo-West Memphis 500kV Line Tap
J1821	Battery Storage	Entergy Arkansas LLC	Lee	AR	NRIS	0	Entergy Keo -West Memphis 500kV Line tap
J1827	Solar	Entergy Louisiana LLC	Morehouse	LA	NRIS	90	Bastrop tap point
J1832	Solar	CLECO Corporation	St. Landry	LA	NRIS	150	Plaisance 138kV
J1833	Gas	Entergy Louisiana LLC	Lafourche	LA	NRIS	115.56	Leeville 115kV
J1834	Gas	Entergy Texas Inc	Orange	TX	NRIS	1230	Sabine 138/230 kV Substation
J1841	Wind	Entergy Arkansas LLC	Cross	AR	NRIS	52	Forrest City to Wynne S 161 KV Line
J1842	Wind	Entergy Arkansas LLC	Cross	AR	NRIS	175	Forrest City to Wynne S 161 KV



1.2. Total Network Upgrades

The cost allocation of Network Upgrades for the projects in the DPP 2020 Cycle 1 South Phase I reflects responsibilities for mitigating system impacts. The total cost of Network Upgrades is listed in Table 2 below. The costs for Network Upgrades are planning-level estimates and subject to revision in the facility studies. Details pertaining to the cost allocation has been detailed in Appendix A – Cost Allocation Summary (CEII).

Table 2: Total Cost of Network Upgrades for DPP 2020 Cycle 1 South Phase I Projects

Project	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrades (\$)	Total Network Upgrade Cost for Milestone Calculation (\$)	M2 (\$)	M3 (\$)
	Thermal	Stability	Short Circuit	Affected System	Deliverability	TO Network Upgrades	TO – Owned Direct Assigned				
a	b	c	d	e	f	g	h	i	j = b + c + d + f + g + i	\$4,000/MW	10% of (j) from Phase I-M2
J1499	-	TBD	TBD	TBD	2,841,559	4,250,000		-	7,091,559	200,000	509,156
J1500	-	TBD	TBD	TBD	14,530,139	2,165,731		-	16,695,870	720,000	949,587
J1509	-	TBD	TBD	TBD	30,436,402	4,250,000		-	34,686,402	400,000	3,068,640
J1514	-	TBD	TBD	TBD	356,854	-		-	356,854	400,000	-
J1523	-	TBD	TBD	TBD	15,203,180	4,250,000		-	19,453,180	200,000	1,745,318
J1539	62,249,229	TBD	TBD	TBD	24,473,642	4,000,000		-	90,722,871	584,000	8,488,287
J1542	3,171,196	TBD	TBD	TBD	139,369,327	16,000,000		-	158,540,522	800,000	15,054,052
J1558	-	TBD	TBD	TBD	9,050,892	16,000,000		-	25,050,892	800,000	1,705,089
J1559	-	TBD	TBD	TBD	13,021,017	4,000,000		-	17,021,017	800,000	902,102
J1560	-	TBD	TBD	TBD	65,043,736	3,150,000		-	68,193,736	800,000	6,019,374
J1561	-	TBD	TBD	TBD	38,862,650	6,500,000		-	45,362,650	600,000	3,936,265
J1562	-	TBD	TBD	TBD	2,185,654	16,000,000		-	18,185,654	800,000	1,018,565
J1564	-	TBD	TBD	TBD	20,121,879	18,000,000		3,748,938	41,870,817	400,000	3,787,082
J1576	-	TBD	TBD	TBD	42,204,401	16,000,000		8,613,054	66,817,455	800,000	5,881,746
J1577	-	TBD	TBD	TBD	3,361,070	4,250,000		-	7,611,070	400,000	361,107
J1583	7,783,034	TBD	TBD	TBD	32,514,673	8,500,000		5,036,739	53,834,445	800,000	4,583,445

J1592	-	TBD	TBD	TBD	9,660,348	16,000,000		-	25,660,348	800,000	1,766,035
J1598	-	TBD	TBD	TBD	126,722,211	16,000,000		-	142,722,211	1,300,000	12,972,221
J1599	-	TBD	TBD	TBD	23,832,262	11,900,000		-	35,732,262	400,000	3,173,226
J1602	-	TBD	TBD	TBD	1,515,450	500,000		-	2,015,450	100,000	101,545
J1607	-	TBD	TBD	TBD	-	4,250,000		-	4,250,000	-	425,000
J1612	-	TBD	TBD	TBD	5,685,699	4,250,000		-	9,935,699	400,000	593,570
J1618	4,361,198	TBD	TBD	TBD	87,662,768	16,000,000		-	108,023,966	800,000	10,002,397
J1630	11,825,131	TBD	TBD	TBD	52,759,799	4,250,000		-	68,834,930	600,000	6,283,493
J1633	-	TBD	TBD	TBD	-	6,550,000		-	6,550,000	-	655,000
J1634	-	TBD	TBD	TBD	36,002,155	6,550,000		-	42,552,155	400,000	3,855,216
J1644	18,937,569	TBD	TBD	TBD	105,896,678	16,000,000		-	140,834,248	600,000	13,483,425
J1668	6,884,573	TBD	TBD	TBD	41,812,364	8,500,000		4,383,926	61,580,862	776,000	5,382,086
J1669	12,000,000	TBD	TBD	TBD	-	16,000,000		-	28,000,000	1,200,000	1,600,000
J1670	-	TBD	TBD	TBD	575,789	-		-	575,789	200,000	-
J1671	-	TBD	TBD	TBD	-	17,500,000		-	17,500,000	600,000	1,150,000
J1672	23,875,573	TBD	TBD	TBD	9,160,500	4,500,000		-	37,536,072	600,000	3,153,607
J1673	-	TBD	TBD	TBD	24,540,229	6,550,000		-	31,090,229	800,000	2,309,023
J1686	5,755,638	TBD	TBD	TBD	70,956,117	11,900,000		-	88,611,755	400,000	8,461,176
J1709	-	TBD	TBD	TBD	115,314,214	4,500,000		-	119,814,214	400,000	11,581,421
J1710	-	TBD	TBD	TBD	3,604,220	4,000,000		-	7,604,220	1,200,000	-
J1722	-	TBD	TBD	TBD	15,362,010	16,000,000		-	31,362,010	400,000	2,736,201
J1723	1,920,233	TBD	TBD	TBD	3,394,103	500,000		-	5,814,336	40,000	541,434
J1747	5,538,697	TBD	TBD	TBD	2,556,692	-		-	8,095,389	200,000	609,539
J1748	-	TBD	TBD	TBD	367,768	750,000		-	1,117,768	400,000	-
J1749	-	TBD	TBD	TBD	35,197,066	4,500,000		-	39,697,066	200,000	3,769,707
J1759	-	TBD	TBD	TBD	-	6,550,000		-	6,550,000	-	655,000



J1760	-	TBD	TBD	TBD	80,661,059	4,500,000		16,452,745	101,613,804	1,000,000	9,161,380
J1789	-	TBD	TBD	TBD	3,863,546	500,000		-	4,363,546	23,960	412,395
J1790	-	TBD	TBD	TBD	-	500,000		-	500,000	-	50,000
J1791	9,460,600	TBD	TBD	TBD	-	4,250,000		-	13,710,600	-	1,371,060
J1794	-	TBD	TBD	TBD	-	17,000,000		-	17,000,000	600,000	1,100,000
J1795	-	TBD	TBD	TBD	-	16,000,000		-	16,000,000	600,000	1,000,000
J1796	-	TBD	TBD	TBD	102,730,499	13,100,000		-	115,830,499	600,000	10,983,050
J1797	-	TBD	TBD	TBD	67,505,396	18,000,000		-	85,505,396	400,000	8,150,540
J1811	-	TBD	TBD	TBD	5,635,564	10,000,000		-	15,635,564	1,000,000	563,556
J1812	41,500,000	TBD	TBD	TBD	101,727,242	6,500,000		-	149,727,242	1,177,000	13,795,724
J1813	-	TBD	TBD	TBD	557,638	TBD		-	557,638	612,000	-
J1816	-	TBD	TBD	TBD	25,714,634	16,000,000		-	41,714,634	540,000	3,631,463
J1818	-	TBD	TBD	TBD	20,691,512	609,334		-	21,300,846	320,000	1,810,085
J1819	-	TBD	TBD	TBD	-	4,000,000		-	4,000,000	-	400,000
J1820	-	TBD	TBD	TBD	1,201,421	4,000,000		-	5,201,421	400,000	120,142
J1821	-	TBD	TBD	TBD	-	4,000,000		-	4,000,000	-	400,000
J1827	-	TBD	TBD	TBD	6,275,754	16,000,000		-	22,275,754	360,000	1,867,575
J1832	13,457,563	TBD	TBD	TBD	121,218,614	3,150,000		-	137,826,177	600,000	13,182,618
J1833	-	TBD	TBD	TBD	71,737,757	4,000,000		-	75,737,757	462,240	7,111,536
J1834	220,500,000	TBD	TBD	TBD	517,948,288	-		-	738,448,288	5,168,000	68,676,829
J1841	11,280,512	TBD	TBD	TBD	17,510,516	4,250,000		-	33,041,028	208,000	3,096,103
J1842	36,099,255	TBD	TBD	TBD	58,944,280	4,250,000		-	99,293,534	700,000	9,229,353

Analyses performed demonstrate that transmission facilities are required to reliably interconnect this group of generators to the transmission system. ERS and NRS Network Upgrades required for full interconnection service detailed in Table 3. Two upgrades, Franklin - McKnight 500kV Line Upgrade and Layfield - Hartsburg 500kV Line Upgrade, are stability limited. Since no dynamic stability model is developed in phase 1, these



upgrades do not have a cost estimate in phase 1 and will have an appropriate network upgrade developed in phase 2 if the constraints continue to exist.

Table 3: ERIS & NRIS Upgrades Required for Interconnection Service

Network Upgrade	Type	TO	NU Cost Estimate (\$)	Projects Impacted
3rd Sabine - Pt. Neches 138 kV line	ERIS	Entergy	\$ 37,000,000.00	J1834
Add 2nd Sabine 230/138 kV autotransformer	ERIS	Entergy	\$ 17,500,000.00	J1834
Brooks Creek - Bayshore 138 kV Line Upgrade	ERIS	Entergy	\$ 17,100,000.00	J1812
Champagne - East Opelousas 138 kV Line Upgrade	ERIS	Entergy	\$ 17,700,000.00	J1539, J1644, J1686, J1832
Clinton Industrial - Norrell 230 kV Line Upgrade	ERIS	Entergy	\$ 9,400,000.00	J1630, J1672, J1747, J1791
Colton - Bloomfield 138 kV Line Upgrade	ERIS	Entergy	\$ 32,400,000.00	J1539, J1644, J1686, J1832
Cross County - Wynne East 161 kV Line Upgrade	ERIS	Entergy	\$ 5,300,000.00	J1723, J1841, J1842
Devin - Wilbert 138 kV Line Upgrade	ERIS	Entergy	\$ 5,300,000.00	J1539, J1644
East Opelousas - Colton 138 kV Line Upgrade	ERIS	Entergy	\$ 8,200,000.00	J1539, J1644, J1686, J1832
Flatland - MidCounty 138 kV Line Upgrade	ERIS	Entergy	\$ 6,500,000.00	J1834
Forest City North - Moses 161 kV Line Upgrade	ERIS	Entergy	\$ 5,600,000.00	J1723, J1841, J1842
J1583 POI - Evergreen 230 kV Line Upgrade	ERIS	Entergy	\$ 19,600,000.00	J1542, J1583, J1618, J1668
J1842 POI - Forest City North 161 kV Line Upgrade	ERIS	Entergy	\$ 10,500,000.00	J1723, J1841, J1842
J1842 POI - Wynne South 161 kV Line Upgrade	ERIS	Entergy	\$ 10,500,000.00	J1841, J1842
New Hartburg - Nelson 230 kV line	ERIS	Entergy	\$ 145,000,000.00	J1834
Pintail - White Bayou Tap 138 kV Line Upgrade	ERIS	Entergy	\$ 14,900,000.00	J1812

Pt. Neches - Dupont Dee 69 kV Line Upgrade	ERIS	Entergy	\$ 12,000,000.00	J1669
Pt. Neches Bulk - Flatland 138 kV Line Upgrade	ERIS	Entergy	\$ 1,500,000.00	J1834
Ray Braswell - Clinton Industrial 230 kV Line Upgrade	ERIS	Entergy	\$ 18,100,000.00	J1630, J1672, J1747, J1791
Replace LA Station 138/69 kV autotransformer and	ERIS	Entergy	\$ 17,500,000.00	J1539
Sabine - Mud Lake 230 kV Line Upgrade	ERIS	Entergy	\$ 1,000,000.00	J1834
Sabine - Oilla 138 kV Line Upgrade	ERIS	Entergy	\$ 12,000,000.00	J1834
Shiloh - Brooks Creek 138 kV Line Upgrade	ERIS	Entergy	\$ 9,500,000.00	J1812
Tinnin - Norrell 230 kV Line Upgrade	ERIS	Entergy	\$ 23,200,000.00	J1630, J1672, J1747, J1791
Wilbert - Addis 138 kV Line Upgrade	ERIS	Entergy	\$ 19,300,000.00	J1539
Willow Glen - Pt. Pleasant 230 kV Line Upgrade	ERIS	Entergy	\$ 1,300,000.00	J1583, J1668
Willow Glen - Wise 230 kV Line Upgrade	ERIS	Entergy	\$ 1,300,000.00	J1583, J1668
Wynne East - Parkin 161 kV Line Upgrade	ERIS	Entergy	\$ 17,000,000.00	J1723, J1841, J1842
Wynne South - Cross County 161 kV Line Upgrade	ERIS	Entergy	\$ 400,000.00	J1841, J1842
2nd J1583 - Evergreen 230 kV Line	NRIS	Entergy	\$ 20,600,000.00	J1542, J1560, J1564, J1583, J1599, J1618, J1668, J1797, J1833
2nd J1630 - Tinnin 230 kV line	NRIS	Entergy	\$ 28,700,000.00	J1509, J1523, J1630, J1672, J1827
2nd Rork - Acadia auto	NRIS	CLECO	\$ 8,000,000.00	J1561, J1598, J1634, J1709, J1760, J1812, J1834
2nd Tunica - J1440 POI 115 kV Line	NRIS	Entergy	\$ 35,500,000.00	J1509, J1523
2nd White Bluff - Keo 500 kV Line	NRIS	Entergy	\$ 100,000,000.00	J1602, J1644, J1686, J1499, J1500, J1509, J1523, J1539, J1542, J1558, J1559, J1560, J1561, J1564, J1576, J1577, J1583, J1592, J1598, J1599, J1612, J1618, J1630, J1634, J1668, J1672, J1673, J1709, J1722, J1747, J1749, J1760, J1796, J1797, J1812, J1818, J1827, J1832, J1833, J1834
3rd Auto Wells	NRIS	CLECO	\$ 8,000,000.00	J1560, J1561, J1598, J1599, J1634, J1644, J1673, J1686, J1709, J1760, J1796, J1812, J1818, J1832, J1834

Adams Creek - Bogalusa 230 kV ckt #1 Line Upgrade	NRIS	Entergy	\$ 2,100,000.00	J1576
Adams Creek - Bogalusa 230 kV ckt #2 Line Upgrade	NRIS	Entergy	\$ 2,100,000.00	J1576
Add 2nd Andrus 230/115 kV autotransformer	NRIS	Entergy	\$ 20,000,000.00	J1630
Add 2nd Nelson 500/230 kV autotransformer	NRIS	Entergy	\$ 20,000,000.00	J1598, J1634, J1673, J1709, J1760, J1812, J1834
Add redundant bus tie breaker at Ritchie 230 kV	NRIS	Entergy	\$ 3,000,000.00	J1509, J1523, J1722, J1789
Barkers Corner - Ramsey Reconductor	NRIS	CLECO	\$ 13,041,600.00	J1564, J1576
Batesville Bus Uprate	NRIS	Cooperative	\$ 1,460,000.00	J1509, J1523, J1789
Bayou Labutte - Willow Glen 500 kV	NRIS	Entergy	\$ 23,500,000.00	J1539, J1560, J1561, J1583, J1598, J1599, J1634, J1644, J1668, J1673, J1686, J1709, J1749, J1760, J1796, J1812, J1818, J1832, J1834
Bayou Vista - Caneland Reconductor	NRIS	CLECO	\$ 59,463,000.00	J1634, J1644, J1560, J1561, J1598, J1599, J1673, J1686, J1709, J1760, J1796, J1812, J1818, J1832, J1834
Bayou Vista - Terebonne Reconductor	NRIS	CLECO/Entergy	\$ 123,880,000.00	J1560, J1561, J1598, J1599, J1634, J1644, J1673, J1686, J1709, J1760, J1796, J1812, J1818, J1832, J1834
Big Cajun - Cocodrie New Line	NRIS	CLECO	\$ 260,400,000.00	J1561, J1598, J1634, J1644, J1673, J1686, J1709, J1760, J1796, J1812, J1818, J1832, J1834
Big Cajun - Fancy Point 500 kV Line Upgrade	NRIS	Entergy	\$ 24,200,000.00	J1560, J1561, J1583, J1598, J1599, J1634, J1644, J1668, J1673, J1686, J1709, J1760, J1796, J1812, J1818, J1832, J1834
Big Cajun - Waterloo 230 kV Line Upgrade	NRIS	Entergy	\$ 4,700,000.00	J1644, J1686, J1749, J1796, J1832
Bloomfield - Vatican 138 kV Line Upgrade	NRIS	Entergy	\$ 1,400,000.00	J1539, J1644, J1686, J1796, J1832
Bobcat - Krotz Springs 138 kV Line Upgrade	NRIS	Entergy	\$ 26,000,000.00	J1560, J1561, J1598, J1599, J1634, J1644, J1686, J1709, J1760, J1796, J1812, J1818, J1832, J1834
Bogalusa - Barkers Corner Reconductor	NRIS	CLECO/Entergy	\$ 40,790,240.00	J1564, J1576
Chalk Bluff - West Memphis 500 kV Line Upgrade	NRIS	Entergy	\$ 2,000,000.00	J1499, J1539, J1542, J1558, J1559, J1560, J1561, J1564, J1576, J1577, J1583, J1592, J1598, J1599, J1602, J1612, J1618, J1630, J1634, J1644, J1668, J1672, J1673, J1686, J1709, J1710, J1722, J1747, J1749, J1760, J1796, J1797, J1811, J1812, J1818, J1820, J1827, J1832, J1833, J1834
Champagne - Bobcat 138 kV Line Upgrade	NRIS	Entergy	\$ 10,100,000.00	J1560, J1561, J1598, J1599, J1634, J1644, J1686, J1709, J1760, J1796, J1812, J1818, J1832, J1834
Cherry Valley - Parkin 161 kV Line Upgrade	NRIS	Entergy	\$ 21,800,000.00	J1811, J1816
Cocodrie - Ville Platte Reconductor	NRIS	CLECO	\$ 19,956,640.00	J1598, J1634, J1644, J1673, J1686, J1709, J1760, J1796, J1812, J1832, J1834

El Dorado - Mt. Olive 500 kV Line Upgrade	NRIS	Entergy	\$ 2,000,000.00	J1499, J1514, J1558, J1559, J1562, J1577, J1592, J1602, J1612, J1670, J1710, J1722, J1723, J1748, J1789, J1811, J1813, J1816, J1820, J1827, J1841, J1842
Fairview - Madisonville 230 kV Line Upgrade	NRIS	Entergy	\$ 1,000,000.00	J1542, J1560, J1583, J1618, J1668, J1797, J1833
Forest City South - Hughes 161 kV Line Upgrade	NRIS	Entergy	\$ 28,500,000.00	J1723, J1789, J1841, J1842
Forest City West - Forest City South 161 kV Line Upgrade	NRIS	Entergy	\$ 3,600,000.00	J1723, J1789, J1841, J1842
Franklin - Bogalusa 500 kV Line Upgrade	NRIS	Entergy	\$ 20,000,000.00	J1499, J1509, J1523, J1602, J1612, J1630, J1672, J1747, J1827
Franklin - McKnight 500kV Line Upgrade	NRIS	Entergy	TBD	J1500, J1539, J1542, J1560, J1561, J1583, J1598, J1599, J1618, J1634, J1644, J1668, J1673, J1686, J1709, J1749, J1760, J1796, J1797, J1812, J1818, J1832, J1833, J1834
Fresh Air - Addis 230 kV Line Upgrade	NRIS	Entergy	\$ 46,100,000.00	J1644, J1686, J1749, J1796, J1832
Frisco - Convent 230 kV Line Upgrade	NRIS	Entergy	\$ 45,200,000.00	J1542, J1618, J1797, J1833
Front Street - Slidel Reconductor	NRIS	CLECO/Entergy	\$ 14,131,200.00	J1539, J1542, J1560, J1561, J1583, J1598, J1599, J1618, J1634, J1644, J1668, J1686, J1749, J1796, J1797, J1818, J1832, J1833, J1834
Gleason - Tyler 161 kV Line Upgrade	NRIS	Entergy	\$ 2,000,000.00	J1558, J1559, J1592
Grimes - Mt. Zion 138 kV Line Upgrade	NRIS	Entergy	\$ 40,200,000.00	J1709
Gypsy - Fairview 230 kV Line Upgrade	NRIS	Entergy	\$ 152,600,000.00	J1542, J1560, J1583, J1618, J1668, J1797, J1833
Hughes - W. Memphis 161 kV Line Upgrade	NRIS	Entergy	\$ 28,500,000.00	J1723, J1789, J1841, J1842
ISES - Powerline Road 500 kV Line Upgrade	NRIS	Entergy	\$ 5,000,000.00	J1796, J1499, J1539, J1542, J1558, J1559, J1560, J1561, J1564, J1576, J1577, J1583, J1592, J1598, J1599, J1602, J1612, J1618, J1634, J1644, J1668, J1673, J1686, J1709, J1710, J1722, J1749, J1760, J1789, J1797, J1811, J1812, J1816, J1818, J1820, J1827, J1832, J1833, J1834
J1368 POI - Fancy Point 230 kV Line Upgrade	NRIS	Entergy	\$ 23,100,000.00	J1644, J1686, J1749, J1796, J1832
J1465 POI - Livonia 138 kV Line Upgrade	NRIS	Entergy	\$ 7,200,000.00	J1560, J1561, J1598, J1599, J1634, J1644, J1686, J1709, J1760, J1796, J1812, J1818, J1832, J1834
J1618 POI - Welcome 230 kV Line Upgrade	NRIS	Entergy	\$ 15,600,000.00	J1542, J1618, J1797, J1833
J1644 POI - Champagne 138 kV Line Upgrade	NRIS	Entergy	\$ 20,500,000.00	J1644, J1686, J1796, J1832

J1668 POI - J1583 POI 230 kV Line Upgrade	NRIS	Entergy	\$ 18,100,000.00	J1542, J1618, J1668, J1797, J1833
J1669 POI - Henning 138 kV Line Upgrade	NRIS	Entergy	\$ 27,200,000.00	J1598, J1634, J1709, J1760, J1812, J1834
J1816 POI - Cherry Valley 161 kV Line Upgrade	NRIS	Entergy	\$ 1,000,000.00	J1811, J1816
Keo - Holland Bottoms 500 kV Line Upgrade	NRIS	Entergy	\$ 3,000,000.00	J1499, J1539, J1542, J1558, J1559, J1560, J1561, J1564, J1576, J1577, J1583, J1592, J1598, J1599, J1602, J1612, J1618, J1630, J1634, J1644, J1668, J1672, J1673, J1686, J1709, J1710, J1722, J1747, J1749, J1760, J1796, J1797, J1812, J1818, J1820, J1827, J1832, J1833, J1834
Keo - J1819 POI 500 kV Line Upgrade	NRIS	Entergy	\$ 2,000,000.00	J1499, J1539, J1542, J1558, J1559, J1560, J1561, J1564, J1576, J1577, J1583, J1592, J1598, J1599, J1602, J1612, J1618, J1630, J1634, J1644, J1668, J1672, J1673, J1686, J1709, J1722, J1747, J1749, J1760, J1796, J1797, J1811, J1812, J1818, J1827, J1832, J1833, J1834
Krotz Springs - Line 642 Tap 138 kV Line Upgrade	NRIS	Entergy	\$ 1,800,000.00	J1560, J1561, J1598, J1599, J1634, J1644, J1686, J1709, J1760, J1796, J1812, J1818, J1832, J1834
Lake Charles Bulk - J1669 POI 138 kV Line Upgrade	NRIS	Entergy	\$ 22,900,000.00	J1598, J1634, J1709, J1760, J1812, J1834
Layfield - Hartsburg 500kV Line Upgrade	NRIS	Entergy	TBD	J1499, J1558, J1559, J1577, J1592, J1602, J1612, J1673, J1710, J1722, J1723, J1789, J1811, J1816, J1820, J1827, J1841, J1842
Line 558/485 Tap - Huntsville 138 kV Line Upgrade	NRIS	Entergy	\$ 7,300,000.00	J1709
Line 637W Tap - Line 658 Tap 69 kV Line Upgrade	NRIS	Entergy	\$ 13,300,000.00	J1644, J1832
Line 642 Tap - J1465 POI 138 kV Line Upgrade	NRIS	Entergy	\$ 33,300,000.00	J1539, J1560, J1561, J1598, J1599, J1634, J1644, J1686, J1709, J1760, J1796, J1812, J1818, J1832, J1834
Line 658 Tap - Sunset 69 kV Line Upgrade	NRIS	Entergy	\$ 18,500,000.00	J1644, J1832
Livonia - Devin 138 kV Line Upgrade	NRIS	Entergy	\$ 47,800,000.00	J1686, J1560, J1561, J1598, J1599, J1634, J1644, J1709, J1760, J1796, J1812, J1818, J1832, J1834
Marked Tree - Harrisburg 161 kV Line Upgrade	NRIS	Entergy	\$ 3,200,000.00	J1710, J1723, J1789, J1816, J1820, J1841, J1842
McKnight - Coly 500 kV Line Upgrade	NRIS	Entergy	\$ 71,900,000.00	J1491, J1492, J1497, J1500, J1509, J1514, J1515, J1516, J1517, J1518, J1523, J1525, J1526, J1527, J1535, J1537, J1550, J1552, J1553, J1557, J1563, J1565, J1571, J1578, J1579, J1586, J1587, J1589, J1591, J1593, J1600, J1601, J1603, J1604, J1609, J1610, J1614, J1616, J1619, J1624, J1625, J1626, J1629, J1630, J1632, J1635, J1637, J1638, J1639, J1641, J1642, J1646, J1649, J1651, J1652, J1653, J1654, J1655, J1656, J1658, J1659, J1660, J1663, J1664, J1665, J1666, J1667, J1672, J1675, J1676, J1677, J1678, J1679, J1680, J1684, J1687, J1690, J1691, J1695, J1696, J1697, J1699, J1701, J1703, J1704, J1706, J1707, J1712,

				J1713, J1714, J1716, J1720, J1721, J1724, J1725, J1726, J1731, J1735, J1736, J1737, J1740, J1743, J1744, J1745, J1746, J1747, J1748, J1755, J1756, J1765, J1770, J1771, J1772, J1777, J1778, J1779, J1780, J1782, J1783, J1784, J1785, J1786, J1793, J1802, J1806, J1810, J1813, J1815, J1824, J1827, J1829, J1830, J1831, J1839, J1840, J1843
Michoud - Front Street 230 kV Line Upgrade	NRIS	CLECO/Entergy	\$ 94,200,000.00	J1539, J1542, J1560, J1561, J1583, J1598, J1599, J1618, J1634, J1644, J1668, J1686, J1749, J1796, J1797, J1818, J1832, J1833, J1834
Morrilton East - Gleason 161 kV Line Upgrade	NRIS	Entergy	\$ 19,500,000.00	J1558, J1559, J1592
Moses - Forest City West 161 kV Line Upgrade	NRIS	Entergy	\$ 8,600,000.00	J1723, J1789, J1841, J1842
Mt. Zion - Line 558/485 Tap 138 kV Line Upgrade	NRIS	Entergy	\$ 14,800,000.00	J1709
Nelson - Richard 500 kV Line Upgrade	NRIS	Entergy	\$ 5,000,000.00	J1499, J1558, J1559, J1577, J1592, J1598, J1602, J1612, J1634, J1673, J1709, J1760, J1812, J1827, J1834
Penton - Nelson Reconductor	NRIS	CLECO/Entergy	\$ 36,944,720.00	J1834, J1598, J1709, J1760, J1812
Penton - Rork New Line	NRIS	CLECO	\$ 198,800,000.00	J1598, J1634, J1709, J1760, J1812, J1834
Perryville - Baxter Wilson 500 kV Line Upgrade	NRIS	Entergy	\$ 21,500,000.00	J1499, J1558, J1559, J1561, J1577, J1592, J1598, J1602, J1612, J1634, J1673, J1709, J1710, J1722, J1723, J1760, J1789, J1796, J1811, J1812, J1816, J1820, J1827, J1834, J1841, J1842
Ponderosa - Conroe 138 kV Line Upgrade	NRIS	Entergy	\$ 10,800,000.00	J1709
Purvis Bulk Transformers Replacement	NRIS	Cooperative	\$ 12,600,000.00	J1500
Replace Brinkley East auto #1 with a larger 161/115 kV auto	NRIS	Entergy	\$ 8,000,000.00	J1722, J1789
Replace Brinkley East auto #2 with a larger 161/115 kV auto	NRIS	Entergy	\$ 8,000,000.00	J1722, J1789
Replace Memphis with a larger 161/500 kV auto	NRIS	Entergy	\$ 18,000,000.00	J1562, J1670, J1723, J1789, J1811, J1816, J1841, J1842
Richard - Wells 500 kV Line Upgrade	NRIS	Entergy	\$ 2,500,000.00	J1598, J1499, J1560, J1561, J1599, J1602, J1612, J1634, J1644, J1673, J1686, J1709, J1760, J1796, J1812, J1827, J1832, J1834
Rivtrin - Staley 138 kV Line Upgrade	NRIS	Entergy	\$ 19,100,000.00	J1709
Rork - Wells New Line	NRIS	CLECO	\$ 75,600,000.00	J1673, J1560, J1561, J1598, J1599, J1634, J1644, J1686, J1709, J1760, J1796, J1812, J1818, J1832, J1834
Sabine - Chisolm Road 230 kV Line Upgrade	NRIS	Entergy	\$ 32,600,000.00	J1709, J1760, J1812, J1834

Sidney - St. James 230 kV Line Upgrade	NRIS	Entergy	\$ 11,200,000.00	J1542, J1797, J1833
St. James - J1618 POI 230 kV Line Upgrade	NRIS	Entergy	\$ 15,600,000.00	J1542, J1797, J1833
Tezcuco - Frisco 230 kV ckt #1 Line Upgrade	NRIS	Entergy	\$ 8,000,000.00	J1542, J1618, J1797, J1833
Tezcuco - Frisco 230 kV ckt #2 Line Upgrade	NRIS	Entergy	\$ 3,500,000.00	J1542, J1618, J1797, J1833
Vacherie - Sidney 230 kV Line Upgrade	NRIS	Entergy	\$ 33,400,000.00	J1542, J1797, J1833
Ville Platte - Westfork Reconductors	NRIS	CLECO	\$ 26,928,000.00	J1598, J1634, J1644, J1673, J1686, J1709, J1760, J1796, J1812, J1832, J1834
Waterford - Tezcuco 230 kV Line Upgrade	NRIS	Entergy	\$ 56,400,000.00	J1542, J1618, J1797, J1833
Waterford Auto - Waterford 230 kV Line Upgrade	NRIS	Entergy	\$ 1,500,000.00	J1542, J1560, J1564, J1583, J1599, J1618, J1668, J1797, J1833
Webre - Bayou Labutte 500 kV Line Upgrade	NRIS	Entergy	\$ 105,700,000.00	J1560, J1561, J1598, J1599, J1634, J1644, J1673, J1686, J1709, J1760, J1796, J1812, J1818, J1832, J1834
Welcome - Bayou Verret 230 kV Line Upgrade	NRIS	Entergy	\$ 15,100,000.00	J1542, J1618, J1797, J1833
Wells - Webre 500 kV Line Upgrade	NRIS	Entergy	\$ 22,000,000.00	J1499, J1539, J1558, J1559, J1560, J1561, J1577, J1592, J1598, J1599, J1602, J1612, J1634, J1644, J1673, J1686, J1709, J1760, J1796, J1812, J1818, J1827, J1832, J1834
Wrightsville - Keo 500 kV Line Upgrade	NRIS	Entergy	\$ 3,000,000.00	J1499, J1539, J1542, J1558, J1559, J1560, J1561, J1564, J1576, J1577, J1583, J1592, J1598, J1599, J1602, J1612, J1618, J1630, J1634, J1644, J1668, J1672, J1673, J1686, J1709, J1722, J1747, J1749, J1760, J1796, J1797, J1812, J1818, J1827, J1832, J1833, J1834

Note:

- 1) Details pertaining to upgrades, costs, and the execution plan for interconnection of the generating facility at the POI will be documented in the Facility Study for Interconnecting Generator.
- 2) Facilities that have been included as base case assumptions and the level of interconnection service that would be conditional upon these facilities being in service will be documented in the GIA (Generator Interconnection Agreement) for each respective GI request successfully achieving GIA execution.
- 3) A detailed summary of the estimated cost of Network Upgrades on a per project basis can be found in Appendix B – Network Upgrade Table Per Project (CEII).

2. FERC Order 827 Compliance Review

The Final Rule of FERC Order 827 “Reactive Power Requirements for Non-Synchronous Generation”, which was issued June 16, 2016, states that “Under this Final Rule, newly interconnecting non-synchronous generators that have not yet executed a Facilities Study Agreement as of the effective date of this Final Rule will be required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.” As such, this Final Rule applies to all wind and solar projects included in the DPP 2020 Cycle 1 South study cycle.

In this study, the power factor at the high-side of the generator substation for each inverter-based project was calculated and reviewed. The study method is to set Qgen of each study project at its Qmax, solve the case, then record the P and Q injection on the high side of the generator substation to calculate the lagging power factor (injecting VAR to the system). The same process is then repeated by setting Qgen at Qmin to calculate the leading power factor (absorbing VAR from the system).

The results show that all projects meet the requirement to maintain 0.95 leading power factor, however, 34 projects do not meet the requirement to provide reactive power capability corresponding to 0.95 lagging power factor, as highlighted in red below in Table 4. Additional reactive support will be needed for these projects to meet the FERC requirement on reactive power capability prior to the completion of their GIA.

Table 4: FERC Order 827 Review Results

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1499	50.5	18.7	N/A	50.2	15.4	0.9560	50.2	-23.2	-0.9077	No	1.1
J1500	150	87.2	N/A	146.5	73.6	0.8936	139.8	-154	-0.6721	Yes	
J1509	102	33.7	N/A	100.2	17.1	0.9857	99.5	-59.5	-0.8583	No	15.8
J1514	104.5	50.6	N/A	102.9	31.3	0.9567	101.8	-87.2	-0.7595	No	2.5
J1523	50.5	19.3	N/A	50.2	17.6	0.9437	50.2	-22.1	-0.9152	Yes	
J1539	147.84	110.88	N/A	144.3	68.2	0.9041	134.6	-272	-0.4435	Yes	
J1542	207.76	90	N/A	203.6	40.7	0.9806	200	-182.7	-0.7383	No	26.2
J1558	204	66.9	N/A	199.8	19.3	0.9954	196.5	-154	-0.7871	No	46.4
J1559	204	66.9	N/A	199.7	17.4	0.9962	196.9	-148.7	-0.7980	No	48.2
J1560	204	66.9	N/A	199.6	17.2	0.9963	196.8	-149.3	-0.7967	No	48.4
J1561	151.2	72.64	N/A	147.5	56	0.9349	145.5	-111	-0.7951	Yes	
J1562	207.76	90	N/A	203.3	35.4	0.9852	198.3	-204	-0.6970	No	31.4
J1564	102	33.7	N/A	100.2	16.9	0.9861	99.5	-58.3	-0.8628	No	16.0
J1576	202.35	66.51	N/A	199	25.2	0.9921	197.6	-128.6	-0.8381	No	40.2
J1577	101.6	55	N/A	100.1	33.3	0.9489	98.3	-106.1	-0.6796	Yes	
J1583	207.76	90	N/A	203.5	39.7	0.9815	200.2	180.2	-0.7433	No	27.2
J1592	204	66.84	N/A	199.7	18.2	0.9959	196.8	-149.9	-0.7955	No	47.4
J1598	332.01	200.001	N/A	327.1	121.7	0.9372	320.6	-389.6	-0.6354	Yes	
J1599	202.4	66.86	N/A	199.6	0.4	1.0000	198.8	-72.1	-0.9401	No	65.2
J1602	27.23	9.1	N/A	27.2	8	0.9594	27.1	-6.9	-0.9691	No	0.9
J1607	25	9.07	N/A	24.9	9.2	0.9380	24.9	-9.1	-0.9392	Yes	
J1612	102	30.8	N/A	101.1	15.6	0.9883	100.5	-63.1	-0.8469	No	17.6
J1618	204	66.88	N/A	199.6	16.4	0.9966	196.4	-155.8	-0.7834	No	49.2
J1630	150	49.3	N/A	147.9	18.3	0.9924	146.8	-96.1	-0.8367	No	30.3
J1633	80	75	N/A	78.6	64.8	0.7716	77.1	-107.4	-0.5832	Yes	
J1634	100	32.87	N/A	98.5	21.3	0.9774	98.1	-51.9	-0.8839	No	11.1
J1644	152.11	67	N/A	149.8	40.4	0.9655	148.3	-110.3	-0.8024	No	8.8
J1668	197.12	147.84	N/A	192.2	89.9	0.9058	184.7	-266.5	-0.5696	Yes	
J1669	166.617	80.696	N/A	164.4	56.5	0.9457	162.5	-140.7	-0.7560	Yes	

Project	Pmax (MW)	Reactive Power Capability (MVar)	Proposed VAR Compensation	VAR Injection			VAR Absorption			Meet FERC Order 827 Requirement?	Add'l VAR Needed (MVar)
				P (MW)	Q (MVar)	Lagging p.f. (pu)	P (MW)	Q (MVar)	Leading p.f. (pu)		
J1670	52.3	25.3	N/A	51.5	14.6	0.9621	50.8	-45.7	-0.7434	No	2.3
J1671	156.8	75.8	N/A	154.4	43.9	0.9619	152.3	-141.1	-0.7336	No	6.8
J1672	156.8	75.8	N/A	152.4	23.3	0.9885	148.3	-142.1	-0.7220	No	26.8
J1673	204.5	95	N/A	200.3	39.6	0.9810	190.9	-254	-0.6008	No	26.2
J1686	101.2	33.437	20 MVAR	102.2	69.3	0.8277	99.5	-101.9	-0.6986	Yes	
J1709	104.94	78	N/A	104	70.6	0.8274	103.3	-91.3	-0.7493	Yes	
J1710	310.3	227.53	87.6 MVAR	283.2	287.3	0.7020	276	-155	-0.8719	Yes	
J1722	101.36	53.25	N/A	100.1	34.6	0.9451	98.9	-92.2	-0.7314	Yes	
J1723	12.4	4.13	N/A	12.4	4	0.9517	12.4	-4.2	-0.9471	No	0.1
J1747	52.3	25.3	N/A	51.6	15.6	0.9572	51.2	-41.6	-0.7761	No	1.4
J1748	111.3	33.5	N/A	109.9	13.5	0.9925	109.5	-60.5	-0.8753	No	22.6
J1749	50	66	N/A	48.1	49.4	0.6976	43.9	-120.2	-0.3431	Yes	
J1759	50	18.15	N/A	49.7	17.9	0.9408	49.7	-19.1	-0.9334	Yes	
J1760	255.99	109	N/A	249.9	47.7	0.9823	243.7	-236.7	-0.7173	No	34.4
J1789	90	29.58	N/A	88.6	16.3	0.9835	88	-48.18	-0.8771	No	12.8
J1790	60	31	N/A	59.5	24.2	0.9263	59.3	-41.3	-0.8206	Yes	
J1791	120	49.3	N/A	118.6	29.2	0.9710	117.9	-79.3	-0.8298	No	9.8
J1794	156.8	75.8	N/A	154.4	44.3	0.9612	152.8	-133.3	-0.7536	No	6.4
J1795	104.544	50.633	N/A	102.9	29.8	0.9605	101.7	-91.5	-0.7434	No	4.0
J1796	150	84	N/A	146.8	51.7	0.9432	144	-146.6	-0.7008	Yes	
J1797	102.06	48.43	N/A	100.8	33.3	0.9495	100.3	-73.4	-0.8070	Yes	
J1811	150	121.08	N/A	147.7	130.6	0.7491	146.9	-125.7	-0.7598	Yes	
J1812	303	150	N/A	296.5	100	0.9476	286	-291	-0.7010	Yes	
J1816	137.6	76.9	N/A	134.9	51.4	0.9345	132.1	-132.1	-0.7071	Yes	
J1818	85.33	36	N/A	84.4	21	0.9704	83.7	-61.5	-0.8059	No	6.7
J1819	159.2	126.4	43.8 MVAR	146.4	181.1	0.6287	143.6	-182.5	-0.6184	Yes	
J1820	103.5	78.99	28.93 MVAR	90.7	105.4	0.6523	89.5	-113.9	-0.6179	Yes	
J1821	53	44.13	14.46 MVAR	41.3	57.3	0.5847	41.3	-52.6	-0.6176	Yes	
J1827	90.2	43.59	N/A	88.5	32.2	0.9397	87.8	-62.8	-0.8134	Yes	
J1832	152.12	73.9	N/A	149.7	42.1	0.9627	147.7	-132.5	-0.7444	No	7.1
J1833	115.56	84.6	N/A	115.2	74.5	0.8397	97.8	-84.2	-0.7578	Yes	
J1834	1320	765.177	N/A	1317	587.5	0.9133	1316.2	-1220.9	-0.7332	Yes	
J1841	56	27	N/A	54.5	18.3	0.9480	53.5	-43.2	-0.7780	Yes	
J1842	179.2	85.26	N/A	175.3	38.5	0.9767	168.1	-165.3	-0.7130	No	19.1

3. Model Development and Study Assumptions

3.1. Base Case Models

The origin of the DPP 2020 Cycle 1 South models is the MTEP20 models with the Bench Cases including all pre-queued projects and associated Network Upgrades, while the Study Cases contain all the interconnection requests in DPP 2020 Cycle 1 South Phase I, in addition to all the facilities in the Bench Cases.

- Bench Cases
 - DPP20-2025SH90-PhaseI-Bench_Final_v2
 - DPP20-2025SUM-PhaseI-Bench_Final_v2
- Study Cases
 - DPP20-2025SH90-PhaseI-Study_Charging_Final_v2
 - DPP20-2025SH90-PhaseI-Study_Final_v2
 - DPP20-2025SUM-PhaseI-Study_Final_v2

3.2. Monitored Elements



Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal rating (PSS®E Rating A), and for NERC category P1-P7 conditions branches were monitored for emergency rating (PSS®E Rating B). Voltage limits were specified for system intact and contingent conditions as per applicable Transmission Owner Planning Criteria.

3.3. Contingencies

The following contingencies were considered in the steady state analysis:

- 1) NERC Category P0 (system intact -- no contingencies)
- 2) NERC Category P1 contingencies
 - a. Single element outages, at buses with a nominal voltage of 68 kV and above
 - b. Multiple element NERC Category P1 contingencies
- 3) NERC Category P2-P7 contingencies
- 4) For all the contingencies and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

3.4. Study Methodology

Non-linear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP 2020 Cycle 1 South generating facilities was evaluated by comparing the steady state performance of the transmission system in the Bench and Study Cases. Analyses used PSS®E version 34.7.0 and TARA version 2001_2.

3.5. Performance Criteria

A branch is considered a thermal constraint if the following conditions are met:

- 1) The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post-contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system-intact condition, or
- 2) The overloaded facility or the overload-causing contingency is at generator's outlet, or
- 3) The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- 4) For any other constrained facility, where none of the Study Generators meet one of the above criteria, however, the cumulative MW impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual MW impact is greater than five percent (5%) of the rating of the facility and has DF greater than five percent (5%) will be responsible for mitigating the cumulative MW impact constraint, or
- 5) Impacts on Affected Systems would be classified as injection constraints based on the Affected Systems' criteria, or
- 6) Any other applicable Transmission Owner Local Planning Criteria are met.

A bus is considered a voltage constraint if both of the following conditions are met:

- 1) The bus voltage is outside of the applicable normal or emergency limits for the post change case, and
- 2) The change in bus voltage is greater than 0.01 per unit

All generators must mitigate thermal injection constraints and voltage constraints in order to obtain any type of Interconnection Service. Further, all generators requesting NRIS must mitigate constraints found by using the Deliverability algorithm to meet the system performance criteria for NERC category P1 events, if DFAX due to the study generator is equal to or greater than 5%.

4. Thermal Analysis

The thermal analysis results for 2020-1 show generator projects J1539, J1542, J1583, J1618, J1630, J1644, J1668, J1669, J1672, J1686, J1723, J1747, J1791, J1812, J1832, J1834, J1841, and J1842 causing constraints. The details pertaining to the thermal analysis can be found in Appendix C – MISO ERIS



Analysis (CEII). No constraints were observed during the voltage analysis.

5. Deliverability Analysis

Generator interconnection projects must pass Generator Deliverability Study to be granted NRIS. If the generator is deemed not fully deliverable, the customer can choose either to change the project to an Energy Resource project, reduce their NR service request amount during decision point, or to proceed with the system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up. The wind and solar generators are tested at 100% of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

The MISO Generator Deliverability Study whitepaper describing the algorithm can be found in BPM 015 – Generation Interconnection, Appendix C.

5.1. Determining the MW Restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study has a DF greater than 5%, part or all its output it is not deliverable. The restricted MW is calculated as following:

$$(MW\ restricted) = \frac{worst\ loading - MW\ rating}{(generator\ sensitivity\ factor)}$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

5.2. Deliverability Study Results

The limiting constraints (monitored facilities – contingency facility pairs) seen in the deliverability analysis for the 2024 Summer case are summarized in Appendix D – MISO Deliverability Analysis (CEII). If a project is not determined to be fully deliverable as a result of the NRIS study, upgrades may be required to attain higher deliverable levels. The NRIS amount evaluated reflects the amount at the generator terminal (reviewed by the interconnection customer during the model review period).

Project	Requested NRIS (MW)	NRIS Amount Evaluated (MW)	Maximum Deliverable Amount (MW)
J1499	50	50	0
J1500	150	150	0
J1509	100	100	0
J1514	100	100	56.74
J1523	50	50	0
J1539	146	146	51.63
J1542	200	200	0
J1558	200	200	70.72
J1559	200	200	70.72
J1560	200	200	0
J1561	150	150	0
J1562	200	200	153.87
J1564	100	100	35.36
J1576	200	200	70.72
J1577	100	100	0
J1583	200	200	70.72
J1592	200	200	70.72

J1598	325	332.01	0
J1599	100	100	0
J1602	25	25	0
J1607	0	0	0
J1612	100	100	0
J1618	200	200	0
J1630	150	150	0
J1633	0	0	0
J1634	100	100	0
J1644	150	150	0
J1668	194	194	68.6
J1669	0	0	0
J1670	50	50	38.47
J1671	0	0	0
J1672	150	150	0
J1673	200	200	0
J1686	100	100	0
J1709	100	100	0
J1710	300	300	25.17
J1722	100	100	0
J1723	10	10	4.21
J1747	50	50	25.19
J1748	100	100	56.74
J1749	50	50	17.68
J1759	0	0	0
J1760	250	250	0
J1789	5.99	5.99	0
J1790	0	0	0
J1791	0	0	0
J1794	0	0	0
J1795	0	0	0
J1796	150	150	0
J1797	100	100	0
J1811	150	150	29.54
J1812	294.25	294.26	0
J1813	107	107	40.81
J1816	135	135	56.9
J1818	80	80	0
J1819	0	0	0
J1820	100	100	8.39
J1821	0	0	0
J1827	90	90	0
J1832	150	150	0
J1833	115.56	115.56	0
J1834	1230	1320	0
J1841	52	52	0
J1842	175	175	0

6. Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) Analysis, which tests for Network Upgrades driven by higher queued

interconnection projects, was performed for this System Impact Study.

The maximum MW impacts and SNU cost allocations appear in Table 5.

Table 5: List of DPP 2020 Cycle 1 South Area Phase I Projects

Network Upgrades	Project Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility (\$)
Lewis Creek - Porter New Line	DPP 2016-AUG	J1472	123.98738	53,643,861	37,191,116.19
	DPP-2020-1	J1760	54.85		16,452,744.81
Bogalusa - Adams Creek 230 kV Upgrade	DPP-2019-1	J1281	4.71	15,000,000	438,804.61
	DPP-2019-1	J1294	7.7616		723,105.28
	DPP-2019-1	J1458	15.844		1,476,097.73
	DPP-2020-1	J1564	40.24		3,748,937.93
	DPP-2020-1	J1576	92.45		8,613,054.45
Bayou Labutte 500 kV Expansion + New Auto + New Line into Wise	DPP-2019-1	J1246	962.18	62,500,000	50,492,285.35
	DPP-2019-1	J1367	9.413775		494,006.33
	DPP-2019-1	J1455	31.005		1,627,048.27
	DPP-2019-1	J1465	8.88		465,995.44
	DPP-2020-1	J1583	95.98		5,036,739.02
	DPP-2020-1	J1668	83.54		4,383,925.58

7. Cost Allocation

The cost allocation of Network Upgrades (NU) for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of NRIS as of the draft System Impact Study report date.

11.1 Cost Assumptions for Network Upgrades

The cost estimate for each NU identified in the System Impact Study is provided by the corresponding Transmission Owner.

11.2 Cost Allocation Methodology

The costs of NU for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Study Case.

Cost Allocation Methodology for Thermal Constraints

1. With all study group generation projects dispatched in the Study Case, all thermal constraints are identified.
2. Distribution factor from each project on each constraint is obtained.
3. For each thermal constraint, the maximum MW contribution (increasing flow) from each project is then calculated in the Post Case without any Network Upgrades.
4. For each thermal constraint, the cost estimates for one or a subset of NU are provided by the corresponding Transmission Owner.
5. The cost of each NU is allocated based on the pro rata share of the MW contribution from each project on the constraints mitigated or partly mitigated by this NU. The methodology to determine



the cost allocation of one NU is:

$$\text{Project A Cost Portion of NU} = \frac{\text{Max(Proj. A MW contribution on constraint)}}{\sum_i \text{Max(Proj. i MW contrution on constraint)}}$$

6. The total NU costs for each project are calculated if more than one NU is required.



Appendix A – Cost Allocation Summary (CEII)

Appendix B – Network Upgrade Table Per Project (CEII)

Appendix C – MISO ERIS Analysis (CEII)

Appendix D – MISO Deliverability Analysis (CEII)